

Introduction

Sets are the boxes of mathematics, collecting objects together to be used for one purpose or another. For the most part, sets are simple objects, though they come with a few subtleties as we shall see.

Learning Objectives

- Use set-builder notation to define a set. Conversely, determine whether elements belong to a set by using set-builder notation.
- Define and compute the union, intersection, complement, and product of sets.
- Determine whether a mathematical statement is a predicate, proposition, or neither.
- Define when AND, OR, and NOT statements evaluate to true and false.
- Use truth tables to evaluate the truthiness of compound propositions.

Sets

Definition 1

A *set* S is any unordered collection of *well-defined* and *distinct* objects. An object of a set is called an *element* of that set. The set which contains no elements is called the *empty set* and is denoted $\emptyset$.

Here, well-defined means there is no ambiguity as to whether an element belongs to a set or not. For example, I cannot define the set of positive numbers which makes me happy. The standard mathematical notation for writing a set is to enclose objects with braces, for example,

$$\{a, b, c\}, \quad \{\smile, \frown\}, \quad \{\heartsuit, \clubsuit, \diamond, \spadesuit\}, \quad \{\text{cat}, \text{dog}\}, \quad \{1, 2, 3, \dots\}.$$

Almost anything can belong to a set, including things like functions and even other sets! To discuss elements of a set, we use the “ $\in$ ” symbol (read as “in”), or “ $\notin$ ” (read as “not in”). For example,

$$a \in \{a, b, c\} \quad \text{but} \quad \text{cow} \notin \{\text{cat}, \text{dog}\}.$$

Exercise 2

Define the sets

$$A = \{1, 2, 2, 3, 3, 3\}, \quad B = \{3, 1, 2\}, \quad C = \{\{1, 2\}, \{3\}\}.$$

Determine which, if any, of these sets are the same.

Set Builder Notation

Most sets we consider will either be large or infinite, making it impractical to write down each element individually. When the number of elements becomes unruly, we depend on *set builder* notation:

$$\{a : a \text{ satisfies property } P\} \quad (\text{read as “the set of all } a \text{ which satisfy } P”).$$

Using a property P to characterize a set is unambiguous and concise. For example

$$\begin{aligned} \{1, 2, 3, \dots\} &= \left\{ a : \begin{array}{l} a \text{ is a whole number} \\ \text{greater than } 0 \end{array} \right\} \\ \{\heartsuit, \clubsuit, \diamondsuit, \spadesuit\} &= \{x : x \text{ is a card suit} \} \\ \{f, g, h, i, j\} &= \left\{ z : \begin{array}{l} z \text{ is a letter of the} \\ \text{alphabet between } e \text{ and } k \end{array} \right\}. \end{aligned}$$

Exercise 3 The following set S , written in set-builder notation, is finite. Explicitly write out the elements of S .

$$S = \{z : z = x + y, x \in \{1, 2\} \text{ and } y \in \{5, 7\}\}.$$

One of the advantages of set-builder notation is that it allows us to construct sets, even when there is no possibility we can list every element in the set. For example, “the set of all people with brown hair” is a perfectly reasonable finite set, despite the fact that one could not explicitly list more than a handful of elements of this set.

Some special sets which we will see often throughout the course are the following:

- The natural numbers $\mathbb{N} = \{0, 1, 2, 3, \dots\}$,
- The integers $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$,
- The rational numbers $\mathbb{Q} = \{\frac{a}{b} : a, b \in \mathbb{Z}, b \neq 0\}$,
- The real numbers $\mathbb{R}$.

Unfortunately, there is a lot of disagreement amongst mathematicians whether 0 should be a natural number. For most of this course, we will instead use the following to avoid confusion:

- The positive integers $\mathbb{Z}^+ = \{1, 2, 3, \dots\}$,
- The non-negative integers $\mathbb{Z}^* = \{0, 1, 2, \dots\}$.

Operations on Sets

Definition 4

Given two sets S and T , we define

- The *union* $S \cup T$ as the collection of all elements contained in either S or T . In set builder notation, this is

$$S \cup T = \{x : x \in S \text{ or } x \in T\}.$$

- The *intersection* $S \cap T$ as the collection of all elements which are contained in both S and T . In set builder notation, this is

$$S \cap T = \{x : x \in S \text{ and } x \in T\}.$$

- The *difference* $S \setminus T$ is the collection of all elements which are in S and are not in T . In set builder notation, this is

$$S \setminus T = \{x : x \in S \text{ and } x \notin T\}.$$

These operations are visualized in terms of the Venn diagrams illustrated in Figure 1

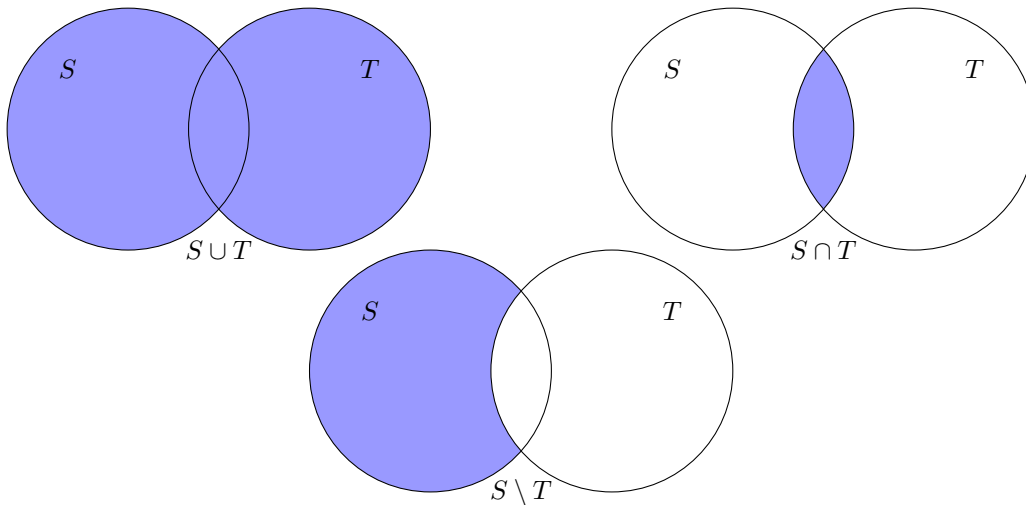


Figure 1: Top Left: The union of two sets is the collection of all elements which are in at least one of the sets. Top Right: The intersection of two sets is the collection of elements which are simultaneously in both sets. Bottom: A set difference consists of all elements in one set that are not elements of the other.

To demonstrate, let $S = \{a, b, c, d\}$ and $T = \{a, b, 1, 2, 3\}$, so that

Union	$S \cup T = \{a, b, c, d, 1, 2, 3\},$
Intersection	$S \cap T = \{a, b\},$
Difference	$S \setminus T = \{c, d\}.$

Example 5

Let $T = \{\dots, -6, -4, -2, 0, 2, 4, 6, \dots\}$ and $S = \{0, 1, 2, 3, 4, \dots\}$. Find a set-builder expression for S and T , then compute and find set-builder expressions for $S \cup T$.

Solution. The set T is the collection of all even numbers, which we will write as

$$T = \{a : a \text{ is an even integer} \}.$$

The set S is similarly just the collection of whole numbers which are greater than or equal to zero, whose set-builder expression is

$$S = \left\{ a : \begin{array}{l} a \text{ is a whole number} \\ \text{greater than or equal to } 0 \end{array} \right\}.$$

The union $S \cup T$ is the collection of all elements which are in either S or T ; that is, we list every element in both sets and omit the duplicates, leaving

$$\begin{aligned} S \cup T &= \{\dots, -6, -4, -2, 0, 1, 2, 3, 4, \dots\} \\ &= \left\{ a : \begin{array}{l} a \text{ is an even number or is a} \\ \text{number greater than or equal to } 0 \end{array} \right\}. \quad \blacksquare \end{aligned}$$

Exercise 6 Repeat Example 5 but now compute $S \cap T$, $S \setminus T$, and $T \setminus S$.

Sometimes when we do mathematics, there is an implicit ambient set called the *universe of discourse*, wherein all of our operations take place. This is necessary to understand the following definition:

Definition 7

Given a universe of discourse U and a set A such that each element of A is also in U , we define the *complement* of A in U to be

$$A^c = \{x \in U : x \notin A\},$$

which is the set of all elements not in A . This is illustrated in Figure 2.

For example, if $U = \mathbb{R}$ is the universe of discourse, and $P = \{x : x > 0\}$ is the set of positive real numbers, then $P^c = \{x : x \leq 0\}$ is the set of non-positive numbers.

Exercise 8 Let $U = \mathbb{Z}$ be the universe of discourse, with S and T defined as in Example 5. Find S^c and T^c .

Definition 9

Given two sets A and B , their *Cartesian product* $A \times B$ is the collection of pairs (a, b) where $a \in A$ and $b \in B$; that is,

$$A \times B = \{(a, b) : a \in A, b \in B\}.$$

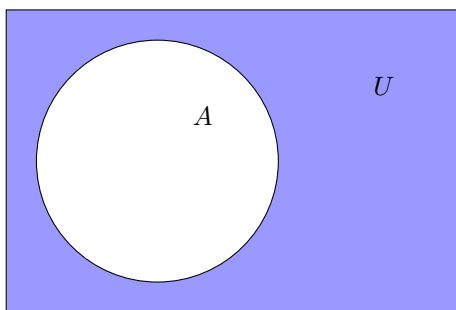


Figure 2: Given a universe of discourse U and a set $A \subseteq U$, the complement of A is the set A^c of elements which are not in A . Here the complement is shaded in blue.

If $S = \{a, b\}$ and $T = \{1, 2, 3\}$ then

$$S \times T = \{(a, 1), (a, 2), (a, 3), (b, 1), (b, 2), (b, 3)\}.$$

Exercise 10 Let $A = \{\heartsuit, \diamondsuit\}$ and $B = \{\uparrow, \downarrow\}$.

1. What is $A \times \emptyset$?
2. Is $A \times B = B \times A$?
3. Is $(A \times B) \times A = A \times (B \times A)$?

Set Relations

Definition 11

Let S and T be two sets. We say that S is a *subset* of T , written $S \subseteq T$, if every element of S is also an element of T .

For example, $\{a, b, c\} \subseteq \{a, b, c, 1, 2\}$ and $\{\text{cat}\} \subseteq \{\text{cat}, \text{dog}, \text{cow}\}$. However, $\{a, b, c\}$ is *not* a subset of $\{a, b, 1, 2\}$ since $c \notin \{a, b, 1, 2\}$.

Example 12

Let A, B be any two sets. Show that $A \subseteq A \cup B$.

Solution. To show that $A \subseteq A \cup B$, we need to show that every element in A is also an element of $A \cup B$. To do this, let $x \in A$ be any element. Since $A \cup B$ is the collection of elements in either A or B , we know that $x \in A \cup B$, showing that $A \subseteq A \cup B$. ■

Exercise 13 Suppose that A and B are sets. Which of the following is true: $A \subseteq A \cap B$ or $A \cap B \subseteq A$? Show that your result is true.

Two sets S and T are equal precisely when they have the same elements. Mathematically, it should be the case that every element of S is in T and vice-versa, so that $S \subseteq T$ and $T \subseteq S$. This serves not only as the mathematical definition of set equality, but tells us how to prove two sets are equal.

Example 14

Consider the sets

$$A = \{n \in \mathbb{Z}^* : n = 4k + 1 \text{ for some } k \in \mathbb{Z}\},$$

$$B = \{n \in \mathbb{Z}^* : n = 4k - 3 \text{ for some } k \in \mathbb{Z}\}.$$

Show that $A = B$.

Solution. Let's begin by showing that $A \subseteq B$. Let $n \in A$, so that there exists a k such that $n = 4k + 1$. Notice that we can equivalently write $n = 4(k + 1) - 3$, showing that $n \in B$. Since n was arbitrary, we conclude that every element in A is also in B , so $A \subseteq B$.

Conversely, if $n \in B$ then $n = 4k - 3$ for some k . We can write this as $n = 4k - 3 = 4(k - 1) + 1$, showing that $n \in A$. Since n was arbitrary, every element of B is also an element of A , so $B \subseteq A$.

Both inclusions imply that $A = B$, as required. ■

Exercise 15 Let A and B be two sets. Show that if $A \subseteq B$ then $A \cup B = B$.

Propositional Logic

Definition 16

A *logical proposition* is a statement which either evaluates to true or false.

Unfortunately, you can see that we've used the word proposition to refer to both to a type of mathematical result, as well as the logical type defined above. While similar, they're treated as being different depending on context. For example, the statements

P_1 : Your mother has green hair.

P_2 : At least 10 people will pass MAT102 this semester.

P_3 : If x and y are real numbers and $x < y$, then $x^2 < y^2$.

are all propositions. Note that propositions do not need to be true, as evidenced by (at least one of) the statements above. However, every proposition is unambiguously either true or false.

Similar to propositions is the notion of a *predicate*. For example, if we define

$$P(x) : x \text{ has green hair,}$$

then we cannot determine whether $P(x)$ is true without specifying a value for x . Here P is the predicate and x is the variable. But predicates can become propositions by specifying a value of the variable, just as $P(\text{your mother})$ is proposition P_1 above.

Exercise 17 Find predicates Q and R such that, if P_2 and P_3 are the propositions above, then $Q(x) = P_2$ and $R(x, y) = P_3$. Be careful regarding P_3 .

There are also statements which are neither predicates nor propositions. For example, “Let $x = \pi$ ” or “How many shirts do you own?” are neither, as they cannot evaluate to true or false.

Do not worry about remembering the differences between propositions and predicates. What’s important is that you are able to identify whether a mathematical statement can be evaluated as true or false; for example, that “ $2 + 2 = 5$ ” is a false statement (a proposition) but “ $x + y = 5$ ” cannot be evaluated without specifying values for x and y (a predicate).

Exercise 18 Identify whether each sentence below is a proposition, a predicate, or neither.

1. Today is Friday.
2. What time is it?
3. The moon is made out of cheese.
4. $x^2 > \pi$.
5. There is an integer solution to the equation $2x^2 - 3y^3 = 4$.
6. $x^2 + y^2 = 1$
7. If there is still food leftover, I’ll have a second helping.

Conjunction, Disjunction, and Negation

We can combine propositions using the three operations listed below.

Definition 19

Suppose that P, Q are predicates, and x, y are fixed values (not variables) so that $P(x)$ and $Q(y)$ are propositions.

- The *conjunction* operator, also known as the AND-operator, is denoted $P(x) \wedge Q(y)$ and pronounced “ $P(x)$ and $Q(y)$.” The statement $P(x) \wedge Q(y)$ evaluates to true when both $P(x)$ and $Q(y)$ are true, and is false otherwise.
- The *disjunction* operator, also known as the OR-operator, is denoted $P(x) \vee Q(y)$ and pronounced “ $P(x)$ or $Q(y)$.” The statement $P(x) \vee Q(y)$ evaluates to true when either or both of $P(x)$ and $Q(y)$ are true, and is false otherwise.
- The *negation* operator, also known as the NOT-operator, is denoted $\neg P(x)$ and pronounced “not $P(x)$.” The statement $\neg P(x)$ evaluates to true when $P(x)$ is false and vice versa.

We can conveniently organize the results of Definition 19 by using a *truth table*, which captures all of the possible truth values for the component propositions and their combined state. For example, the AND, OR, and NOT truth tables are provided in Table 1.

AND			OR			NOT	
P	Q	$P \wedge Q$	P	Q	$P \vee Q$	P	$\neg P$
T	T	T	T	T	T	T	F
T	F	F	T	F	T	F	T
F	T	F	F	T	T	T	F
F	F	F	F	F	F	F	T

Table 1: The truth tables for the AND, OR, and NOT operations. Here T indicates True and F indicates False.

For example, suppose we’re given the predicates $P(x)$: “ x is prime” and $E(y)$: “ y is even,” so that

$$E(2) \wedge P(2) \text{ is true, } E(3) \wedge P(5) \text{ is false,}$$

$$E(2) \wedge P(4) \text{ is false, } E(5) \wedge P(12) \text{ is false.}$$

while

$$E(2) \vee P(2) \text{ is true, } E(3) \vee P(5) \text{ is true}$$

$$E(2) \vee P(4) \text{ is true, } E(5) \vee P(12) \text{ is false.}$$

Exercise 20 Suppose that we're working on the subsets of $\mathbb{Z}$ (so that the subsets of $\mathbb{Z}$ form the universe of discourse), and define the predicates

$P(x)$: "The set x has an even integer."

$Q(x)$: "The set x contains more than 3 elements."

Define the sets

$$A = \{x \in \mathbb{Z} : x^2 < 2\} \quad \text{and} \quad B = A \cup \{x \in \mathbb{Z} : x = y - 1, y \in A\}.$$

Determine the truth of each of the following statements:

- | | | |
|---------------------|-----------------------|-----------------------------|
| 1. $\neg P(B)$ | 3. $P(A) \vee Q(B)$ | 5. $(\neg P(B)) \vee Q(A)$ |
| 2. $P(A) \vee Q(A)$ | 4. $Q(A) \wedge Q(B)$ | 6. $\neg(P(A) \wedge Q(B))$ |

Truth tables can be used to determine the truthiness of more complicated expressions, and even to show that two statements are *logically equivalent*. That is to say, if two expressions produce the same truth table, they are the same statement. We will write $P = Q$ to indicate that P and Q are logically equivalent.

Proposition 21

If P and Q are predicates, then $\neg(P \wedge Q) = \neg P \vee \neg Q$.

Proof. It suffices to show that the expressions have equivalent truth tables. The truth tables are below, with some intermediate steps provided for clarity:

P	Q	$P \wedge Q$	$\neg(P \wedge Q)$	P	Q	$\neg P$	$\neg Q$	$(\neg P) \vee (\neg Q)$
T	T	T	F	T	T	F	F	F
T	F	F	T	T	F	F	T	T
F	T	F	T	F	T	T	F	T
F	F	F	T	F	F	T	T	T

The resulting values of the truth table are identical given the same inputs for P and Q . Therefore, these statements are logically equivalent. $\square$

Exercise 22 Suppose that P and Q are predicates. Form a hypothesis as to what should happen when we "distribute" the negation over the expression $\neg(P \vee Q)$. Prove that your hypothesis is correct.

Exercise 23 Suppose that P, Q , and R are predicates. Form the truth table for the predicate $(\neg P \vee Q) \wedge R$.

Exercises

1. Explicitly write down each of the following sets:
 - (a) $\{x \in \mathbb{Z}^+ : 3 < x^2 \leq 80\}$.
 - (b) If $S = \{1, 2, 3\}$ then $\mathcal{P}(S) = \{A : A \subseteq S\}$.
 - (c) If $A = \{-1, 1\}$ and $B = \{0, 2, 4\}$ then $\{z \in \mathbb{Z}^* : z = y - 2x, x \in A \text{ and } y \in B\}$.
 - (d) $\{(x, x^2) : x \in \{-1, 0, 1\}\}$.
2. Consider the sets

$$A = \{n \in \mathbb{Z} : n = 8x + 3y, x, y \in \mathbb{Z}\}, B = \{n \in \mathbb{Z} : n = 5x + 3y, x, y \in \mathbb{Z}\}.$$

Show that $A \subseteq B$. Is $B \subseteq A$?

3. Let U be the universe of discourse with $A, B, C \subseteq U$. Prove each of the following set identities:
 - (a) $A \cap A^c = \emptyset$
 - (b) $A \setminus B = A \cap B^c$
 - (c) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
 - (d) $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$
 - (e) $(A \cap B) \times C = (A \times C) \cap (B \times C)$
4. Construct truth-tables for the following statements:
 - (a) $(P \wedge Q) \vee R$
 - (b) $(P \wedge \neg Q) \vee (Q \wedge \neg R)$
 - (c) $(P \vee R) \wedge (Q \vee R)$
 - (d) $\neg(P \vee Q) \vee (Q \wedge \neg R)$
5. We say that two logical statements are *logically equivalent* if they produce the same truth tables. Which, if any, of the statements from Exercise 4 are logically equivalent?
6. Let $A = \{-2, -1, 0, 1, 2\}$. Define the following predicates on subsets of A :

$P(S)$: S contains at most three elements.

$Q(S)$: S contains exactly one negative number.

$R(S)$: The sum of the elements in S are non-negative.

Explicitly write out all of the sets S which satisfy the statement $P(S) \wedge Q(S) \wedge (\neg R(S))$.